

Nayib Martín Goushesh

Barcelona | [LinkedIn](#) | [Blog](#) | +34 634580566 | nayib3131@gmail.com

EDUCATION

Politechnic University of Catalonia (UPC)

Bachelor of Aerospace Systems Engineering (Air Navigation), EETAC

Universitatea Politehnica din București (UPB)

ERASMUS+, Air Navigation

Castelldefels, Barcelona

Graduation Date: June 2024

Bucharest, Romania

02/2023-06/2023

PROFESSIONAL SUMMARY

Data-driven Aerospace Systems Engineer with expertise in air traffic management, avionics, and related aerospace domains, complemented by advanced programming and data analysis skills. Proven ability to leverage data to optimize aerospace systems, improve efficiency, and solve complex engineering challenges. Passionate about bridging aerospace innovation with data-driven insights to transform technical problems into actionable solutions. Seeking a challenging role at the intersection of aerospace engineering and data science.

EXPERIENCE

Eurofirms - Vueling

AOG Desk Purchasing Engineer

Barcelona

10/2024-Present day

- Part sourcing, and logistics coordination to minimize aircraft downtime, ensuring efficient resolution of grounding issues.

Design Thinking Contest SEAT

Finalist

Barcelona

03/2024-05/2024

- Envisioned the future of an in-vehicle virtual assistant.
- Designed and showcased a MVP based on a *generative AI agentic framework*.

CRIDA (R&D center for ATM innovation)

R&D Engineer

Remote

02/2024-09/2024

- Developed a custom Deep Reinforcement Learning (DRL) framework, encompassing a realistic simulation environment, a reward function, and a robust training process with integrated monitoring mechanisms.
- Applied PPO and DQN algorithms for autonomous drone traffic management within a simulated urban vertiport environment.

Freelance

Developer

Remote

- Engineered a Python script for backtesting stock trading strategies, processing historical data for all tickers within the Russell 1000 Index and calculating key performance metrics to evaluate strategy effectiveness.
- Developed a series of time series prediction and classification models using Machine Learning and Deep Learning architectures (including LSTM, CNN, and Transformer networks) with Keras and TensorFlow, creating custom data pipelines for efficient model training and evaluation.
- Leveraged local Large Language Models – specifically GPT4All and Ollama – in conjunction with Langchain to create AI-powered agents that streamline the YouTube video creation process, from script generation to content summarization, enhancing creative workflows.

SKILLS & INTERESTS

Technical Proficiency: C#, MATLAB, Python, Git, SkyDemon, Proteus, SolidWorks, Ansys Fluent, Microsoft Excel, Keras, Tensorflow, Pytorch, Scikit-learn, stable-baselines3, gymnasium.

Project Management: Team collaboration, design, planning, execution, testing

Analytical Skills: Problem Solving, Data Analysis, Critical Thinking, Research

Communication: Effective oral/written communication, project report preparation

What I did in University (Some projects are detailed in my [Linkedin](#))

Single European Sky ATM Research (300420 - ATM-OA): Studied the SESAR framework and advanced ATM concepts (ConOps, complexity, FABs, trajectory-based operations, synchronization, conflict automation); analyzed research articles and performed data analysis/modeling related to flight data, machine learning, noise/emissions, and conflict resolution. Activities: Lectures, Tasks, Exam, Data Analysis Exercises.

Flight Mechanics (300277 - MV): Applied general equations of motion; analyzed performance of gliders (incl. wind) and jet aircraft (level flight, climb, range, endurance, takeoff/landing); studied longitudinal and lateral-directional static stability/control concepts. Activities: Problem Solving, Directed Activities, Exams.

Aeronautical Communications 2 (300248 - CA2-MN6): Analyzed RF propagation, transmission lines (Smith Chart), and antennas; performed link budget/coverage calculations; studied Terra-Aire systems (HF/VHF analog/digital VDL, CPDLC) and Terra-Terra networks (AFTN evolution to ATN, REDAN, VoIP). Activities: Labs, Problem Solving, In-depth System Analysis Project.

Projects in Air Traffic Management (300249 - PGTA-MN7): Executed three team-based projects on industry challenges: redesigned airspace/procedures (LERS CTR) using PANS-OPS; developed an Asterix surveillance data decoder using C#; analyzed ATM departure trajectories (LEBL) using Python. Utilized collaborative tools (Github) and presented results. Activities: Team Projects, Presentations.

Sustainability of Aerospace Engineering (300247 - SEA): Analyzed global unsustainability causes; applied sustainable development principles and systemic thinking; examined globalization impacts and human values in engineering; utilized sustainability assessment tools (LCA, EIA, CSR standards like GRI/ISO 26000); evaluated the specific economic, environmental, and social impacts of aerospace engineering. Activities: Theoretical/Practical Questionnaires.

Structures & Strength of Materials (300245 - ERM): Calculated section properties (centroids, moments of inertia); applied 2D elasticity theory (stress, strain, Hooke's Law); determined stress resultants (N, V, M) in isostatic structures; calculated normal stresses due to axial load/bending (Navier's formula) and shear stresses due to shear force (Collignon-Jourawski formula). Activities: Problem Solving, Evaluable Sessions, Exams.

Aeronautical Communications 1 (300243 - CA1-MN1): Analyzed communication system blocks, noise sources, antenna parameters, and link budgets; reviewed analog modulations (AM/FM) and systems (ILS/DME/VOR); studied digital baseband/passband transmission (PAM, line codes, matched filter, BER); investigated multiplexing (TDM/FDM/OFDM) and multiple access (TDMA/FDMA/CDMA) techniques. Activities: Problems, Controls, Labs, Group Work.

Models for Air Traffic Management (300244 - MGTA): Applied optimization (LP, GA, Ant Colony) to model runway capacity, sequencing, sectorization, flow programs, and traffic assignment; analyzed ATC automation (complexity metrics, AMAN/DMAN, conflict detection/resolution) and performance indicators (incl. carbon footprint); studied SESAR/NextGen projects. Activities: Theory/Lab Classes, Group Projects, Presentation, Exams.

Further Mathematics 2 (300242 - AM2): Applied combinatorics and probability theory; analyzed discrete/continuous distributions; performed statistical estimation (confidence intervals) and linear regression; utilized Taylor/power series and numerical methods for differentiation and ODE/PDEs; applied graph theory algorithms (MST, shortest path, flows, PageRank). Used Minitab for statistical analysis. Activities: Problems, Controls, Minitab Lab, Directed Activities, Exams.

Air Navigation, Cartography & Cosmography (300229 - NACC-MN4): Studied air navigation fundamentals (VOR, DME, ILS, SatNav); analyzed orbital mechanics and GPS time/observables/errors; derived GPS navigation equations and Dilution of Precision (DOP); examined GNSS receiver architecture; investigated sensor fusion using Kalman filters and IMUs. Activities: Exams, Lab Practices.

Avionics (300228 - AVI-MN2): Analyzed avionic architectures (FbW, EFIS, ECAM, FMS, Autopilot, FWS); studied industrial design/safety processes and standards (ARP4761/4754, DO-178B/254); investigated hardware (ADIRU, data buses: ARINC 429/AFDX), software (RTOS), and integration aspects (EMC). Applied concepts in group projects and simulator visits. Activities: Theory, Problems, Controls, Projects, Visit.

Radiolocation (300227 - RL-MN8): Analyzed pulsed/Doppler/MTI radar principles (equation, detection, clutter); studied components (antennas) and signal processing (matched filter, CFAR, pulse compression); investigated tracking radar, SSR, ADS-B, ACAS, EGPWS, and weather radar systems using analysis and MATLAB simulation. Activities: MATLAB Practices, Controls, Exams.

Electronics (300222 - ELECTRO): Analyzed analog signal acquisition, conditioning, and processing; studied basic nonlinear electronic components (diodes, transistors) and circuits; covered digital signal and system fundamentals; introduced programmable electronic systems (microprocessors/microcontrollers). Activities: Exercises, Controls, Exams, Labs, Project.

Control and Guidance (300225 - CG-MN3): Modeled dynamic systems (Laplace, State-Space); designed analog controllers using Root Locus and Frequency Response (Bode plots); designed digital controllers (Z-transform, deadbeat); applied modern state-space control techniques; designed and implemented basic autopilots using MATLAB/Simulink. Activities: MATLAB Labs, Controls, Exams.

Aerial Operations (300223 - OA-MN5): Reviewed aircraft cockpit instrumentation and systems (CNS/FANS); analyzed aircraft performance limitations (speeds, mass/CG, takeoff/landing distances, payload-range); performed flight planning (fuel calc., Cost Index); studied special operations (Reduced Thrust TO, ETOPS) and contingency procedures (engine failure). Activities: Theory/Practical exercises, Project.

Graphic Expression (300219 - EG): Applied technical drawing standards; created/analyzed complex 3D solid models and assemblies using feature-based CAD (SolidWorks), focusing on design intent; simulated basic mechanisms; generated standardized 2D drawings; interpreted and modified topographic data (cut/fill). Activities: CAD exercises, Directed Activities.

Materials Science & Technology (300215 - CTM): Analyzed mechanical properties (stress-strain, hardness, toughness) and failure modes (fracture, fatigue, creep) of metallic materials; related properties to microstructure, dislocations, and strengthening mechanisms (heat treatments); interpreted phase diagrams (Fe-C) and TTT diagrams; studied aerospace alloys (Al, Ti, superalloys) and composite materials/manufacturing. Performed tensile/hardness tests. Activities: Lectures, Problems, Labs, Controls, Exams.

Linear Systems (300217 - SL): Classified signals/systems; applied Laplace Transforms to analyze LTI system dynamics (transfer functions, pole-zero diagrams, stability, time response); evaluated feedback structures using block diagrams; analyzed frequency responses (Bode plots) and designed basic filters using Fourier methods and asymptotic analysis. Utilized MATLAB/Simulink for analysis/simulation. Activities: Workshops, Controls, Labs, Exams.

Electricity (300214 - ELECTRI): Analyzed DC/AC circuits using Kirchhoff/Ohm's laws and phasors; calculated power/power factor in single/three-phase systems; studied power electronic converters (rectifiers, inverters), transformers (tests, types), batteries, and generators (synchronous, DC); solved problems related to aeronautical electrical systems using analysis and simulation (Proteus). Activities: Problems, Controls, Exams, Labs, Group Work.

Air Transport Infrastructures (300213 - ITA): Analyzed CNS/ATM systems (Comm: CPDLC; Nav: GNSS/SBAS; Surv: SSR/ADS-B/ACAS); studied airspace structure (FIRs, classes) and flight procedures (VFR/IFR, SIDs/STARs, Approaches, RNAV); investigated ATM services (ASM, ATFM, ATS), separation standards (RVSM), and CDM; designed instrument procedures. Activities: Discussions, ATC Practices, Exercises, Group Project.

Informatics II (300212 - I2): Studied modern computer architectures; developed object-oriented visual applications (C#) using advanced data structures (stacks, queues) and algorithms (sorting, search); managed

dynamic memory; designed and queried relational databases (SQL). Activities: Exercises, Group Project, Presentation.

Thermodynamics (300209 - TER): Applied 1st/2nd Laws to closed/open systems; analyzed properties of pure substances and ideal/real gases (incl. kinetic theory); evaluated performance of devices and cycles (Otto, Diesel, Brayton, Rankine, Refrigeration); performed entropy analysis; covered basics of combustion, psychrometrics, and heat transfer mechanisms. Activities: Problem Sets, Controls, Demo Practices.

Fluid Mechanics (300211 - MF): Analyzed fluid kinematics and statics; applied integral/differential forms of conservation laws (mass, momentum, energy), including Bernoulli and Navier-Stokes equations; performed dimensional analysis; investigated incompressible potential flow and compressible flows (Mach number, shocks, nozzles). Applied concepts to aerodynamics. Activities: Problem Sets, Controls, Exams, Numerical Project.

Aerospace Technology & Air Transport (300208 - TA): Surveyed aerospace technology: aircraft parts/types, flight principles, stability/control, performance optimization, materials/structures, propulsion, air navigation systems/instruments, aircraft systems, helicopters, and space systems. Activities: Classes, Practicals, Project.

Mechanics (300206 - MEC): Applied Newton's laws with variable forces and conservation theorems; analyzed oscillations (damped, forced, resonance); solved central force problems (Kepler's laws, orbital transfers); analyzed particle systems (incl. rockets) and rigid body dynamics using moving frames and Euler's equations/angles. Developed numerical integration code. Activities: Problem Sets, Controls, Exams, Computational Project.

Informatics I (300207 - II): Learned OS fundamentals; developed programs using basic elements (data types, control flow, arrays), advanced concepts (structs, files, functions), and basic algorithms; utilized programming environment for project creation, compilation, and debugging. Activities: Exercises, Submissions, Team Project (with TA course), Presentations.

Business (300204 - EMP): Applied Lean Startup methodology using Business/Lean Canvas; analyzed market competition (Porter's 5 Forces), customer segments, and viability; developed Marketing and Financial Plans (incl. basic accounting, investment evaluation); created and presented an Elevator Pitch and final Business Plan. Studied air transport market specifics via a video report. Activities: Team Project, Presentations, Info Search.

Further Mathematics (300205 - AM): Calculated multiple integrals (double, triple) using coordinate changes (polar, spherical, cylindrical); performed line/surface integrals for scalar/vector fields; applied vector calculus theorems (Green, Stokes, Gauss) and the Nabla operator; developed Fourier series (convergence, Parseval) and Fourier transforms (properties, convolution, generalized functions). Activities: Cooperative Sessions/Deliverables, Controls, Exams.

Calculus (Aeronautics) (300203 - CAL-A): Analyzed single-variable functions (limits, continuity, derivatives, Taylor series, optimization); performed integration (techniques, area/volume applications, improper integrals); identified conics/quadratics; applied multivariable differential calculus (partial/directional derivatives, gradient, Jacobian, constrained extrema using Lagrange multipliers). Activities: Controls, Exams.

Fundamentals of Physics (300201 - FF): Applied principles of mechanics (kinematics, Newton's laws, work/energy theorems, conservation laws, rotation); analyzed electrostatics (fields, potential, Gauss's law) and magnetostatics (forces, Ampere's law); studied electromagnetic induction (Faraday's law), mechanical/EM waves, and Maxwell's equations synthesis. Activities: Controls, Exams, Problem Sets.

Chemistry (300202 - QUI): Analyzed states of matter, atomic/molecular structure, bonding, and solid-state properties (crystallography, defects, alloys); applied principles of thermochemistry, kinetics, and equilibrium to reactions (incl. combustion); investigated electrochemistry (redox, batteries) and corrosion processes; studied basic organic chemistry and fuel characteristics. Activities: Theory/Problem Classes, Labs, Controls, Exams, Tasks.

Algebra & Geometry (300200 - AG): Manipulated complex numbers (Euler's); solved linear systems (Gauss, Cramer) using matrices/determinants; analyzed vector spaces, bases, and scalar products; performed linear transformations and matrix diagonalization; solved ordinary differential equations (1st/higher order) analytically and using Laplace transforms (incl. convolution, Dirac delta). Activities: Controls, Exams.
